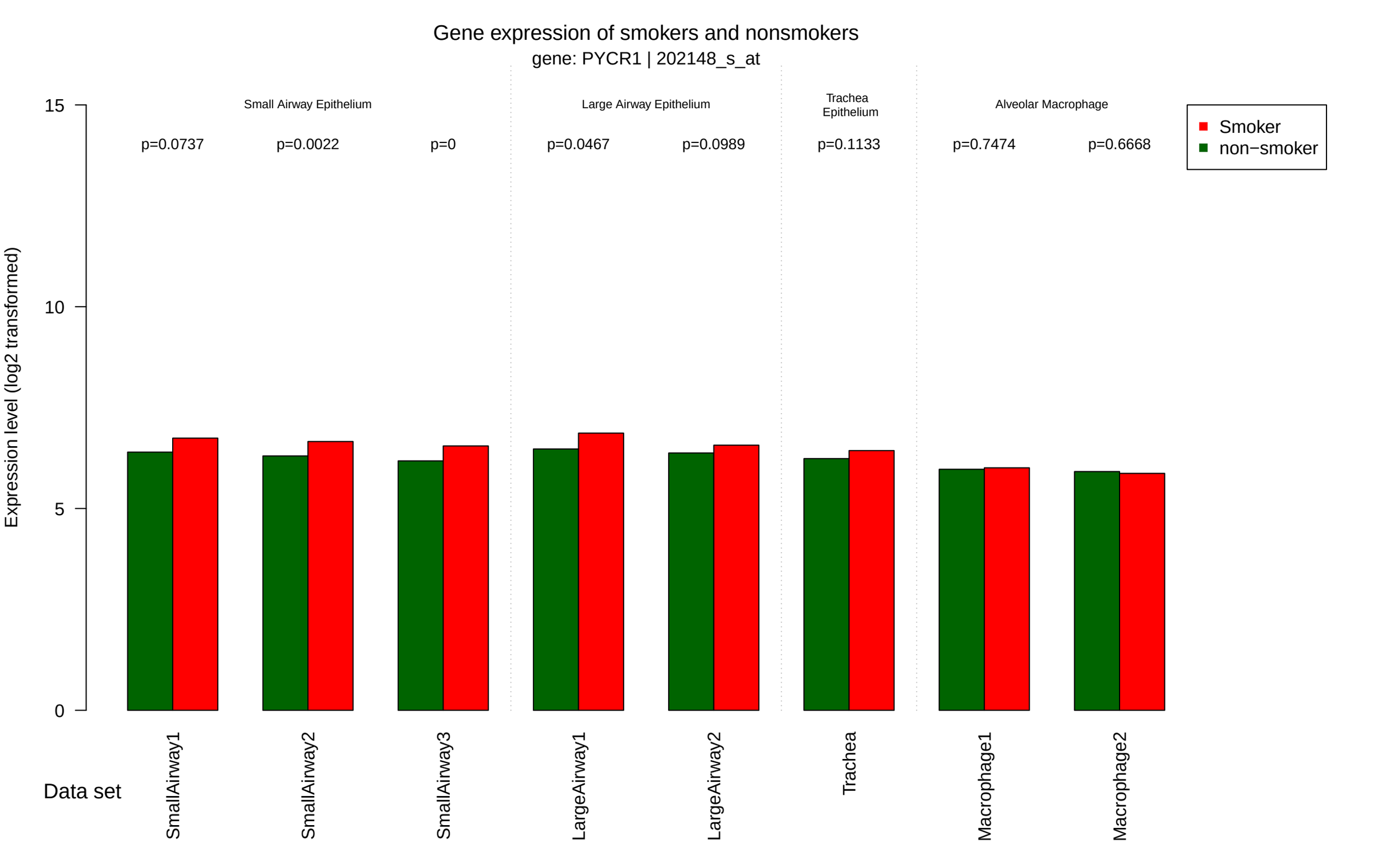
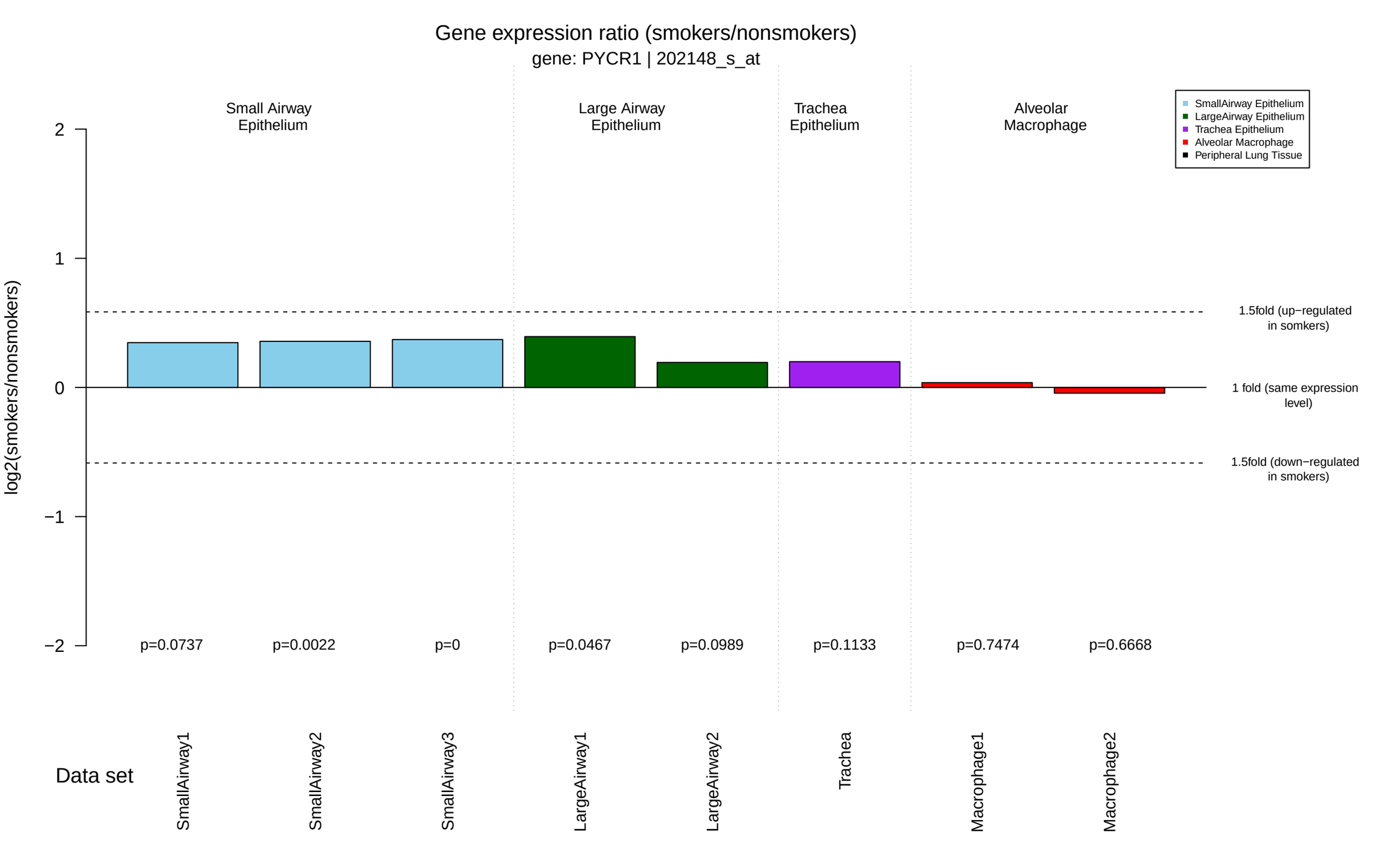
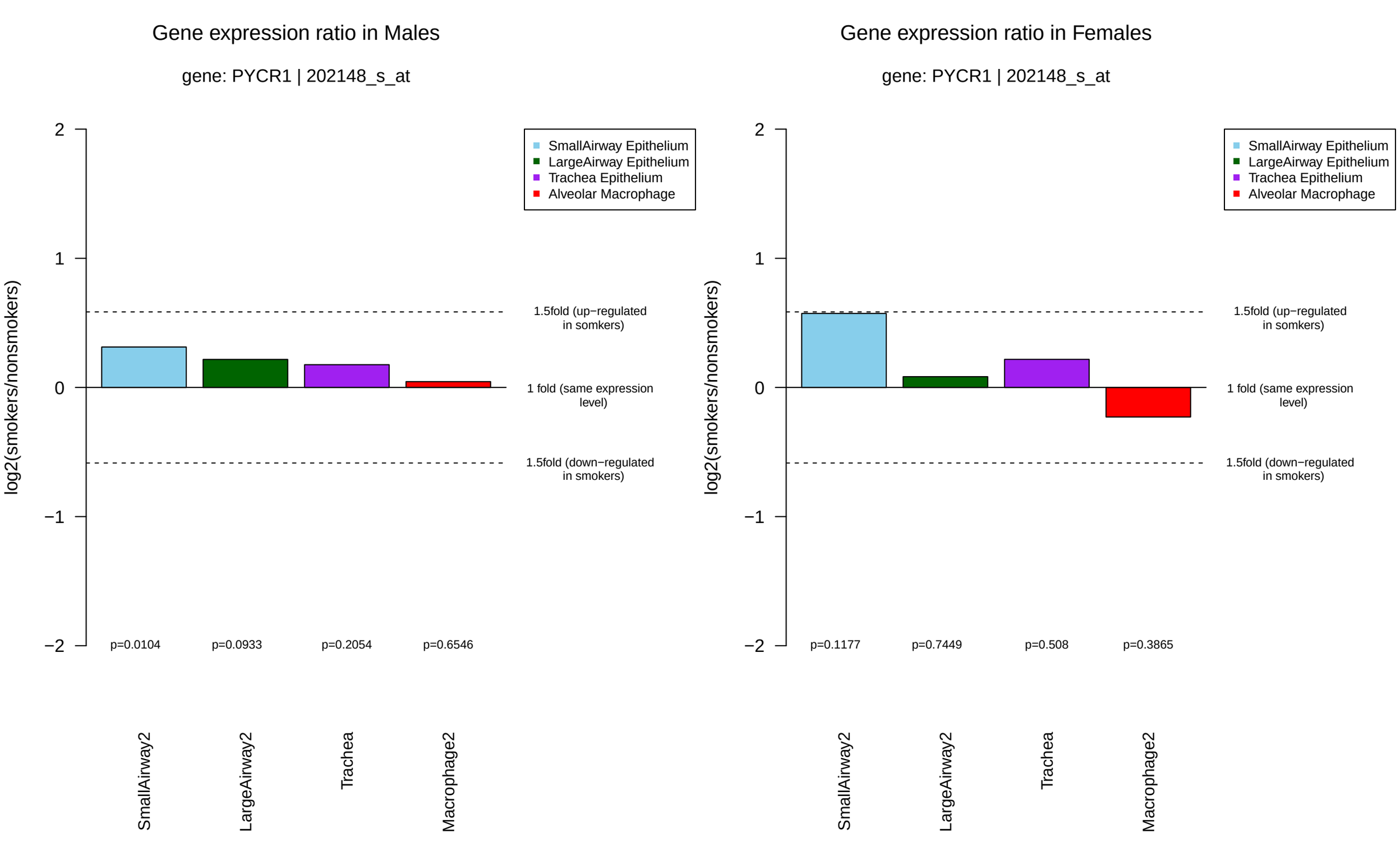
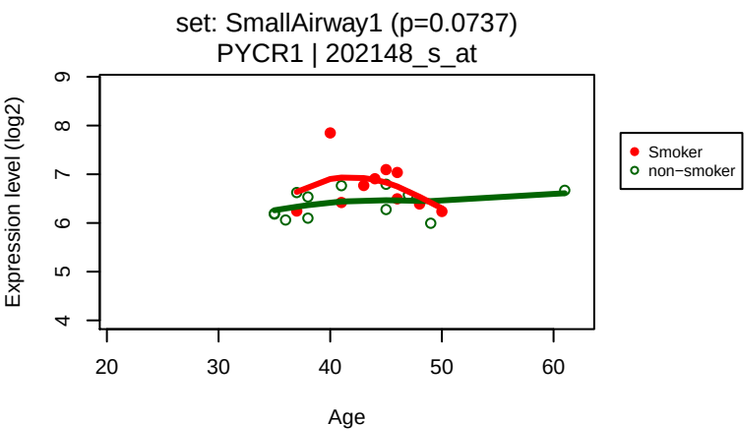
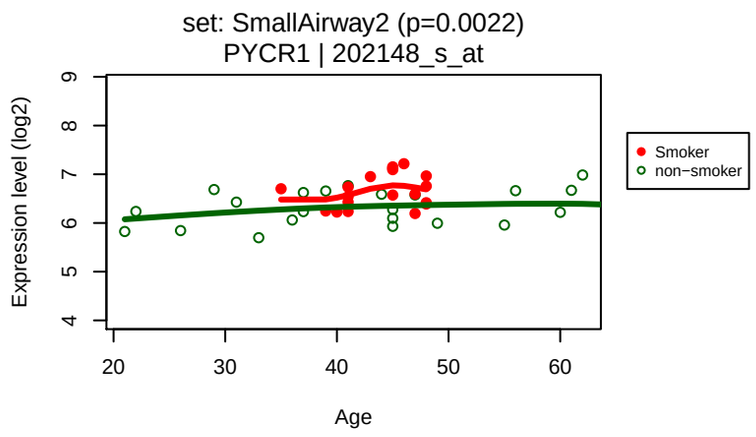
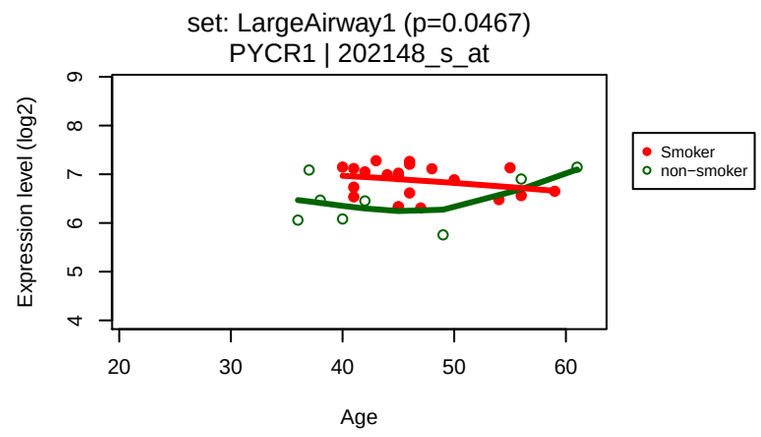

Gene expression of smokers and nonsmokers
gene: PYCR1 | 202148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0737
p=0.0022
p=0
p=0.0467
p=0.0989
p=0.1133
p=0.7474
p=0.6668
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

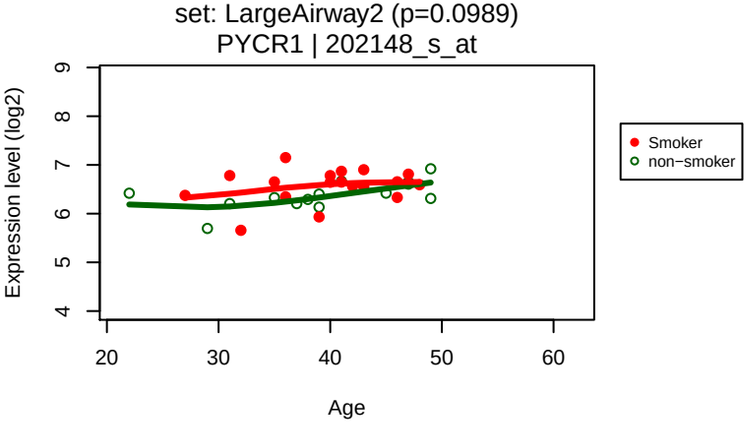

Gene expression ratio (smokers/nonsmokers)
gene: PYCR1 | 202148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0737
p=0.0022
p=0
p=0.0467
p=0.0989
p=0.1133
p=0.7474
p=0.6668
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

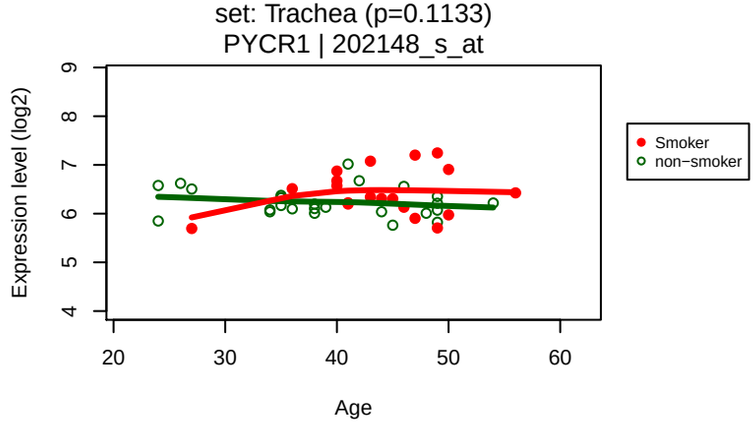

Gene expression ratio in Males
gene: PYCR1 | 202148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0104
p=0.0933
p=0.2054
p=0.6546
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PYCR1 | 202148_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1177
p=0.7449
p=0.508
p=0.3865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

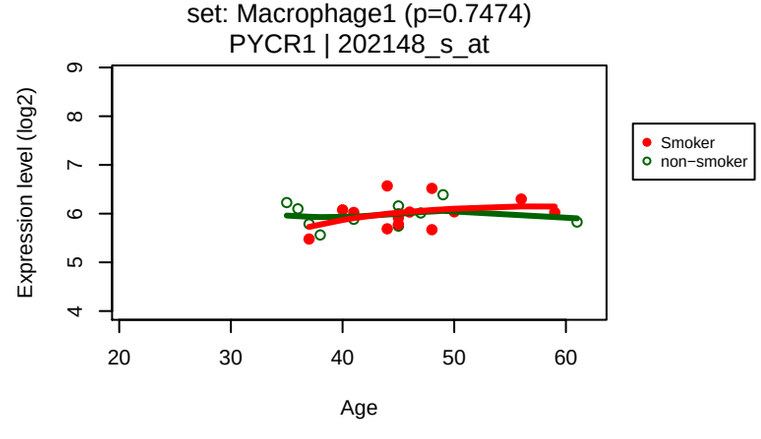

set: SmallAirway1 (p=0.0737)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker

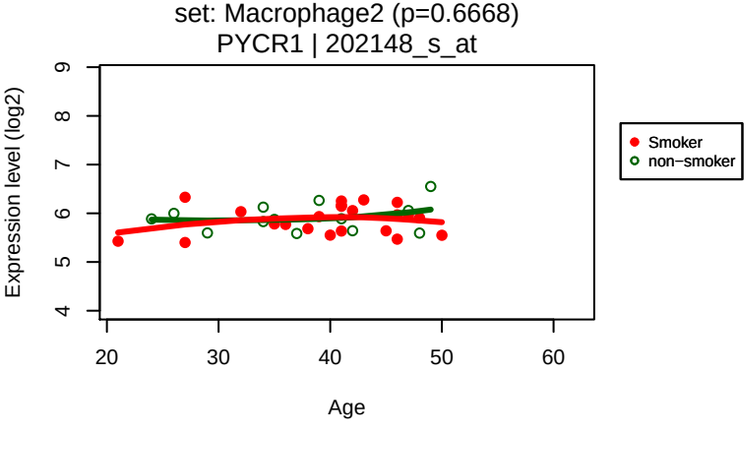

set: SmallAirway2 (p=0.0022)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker

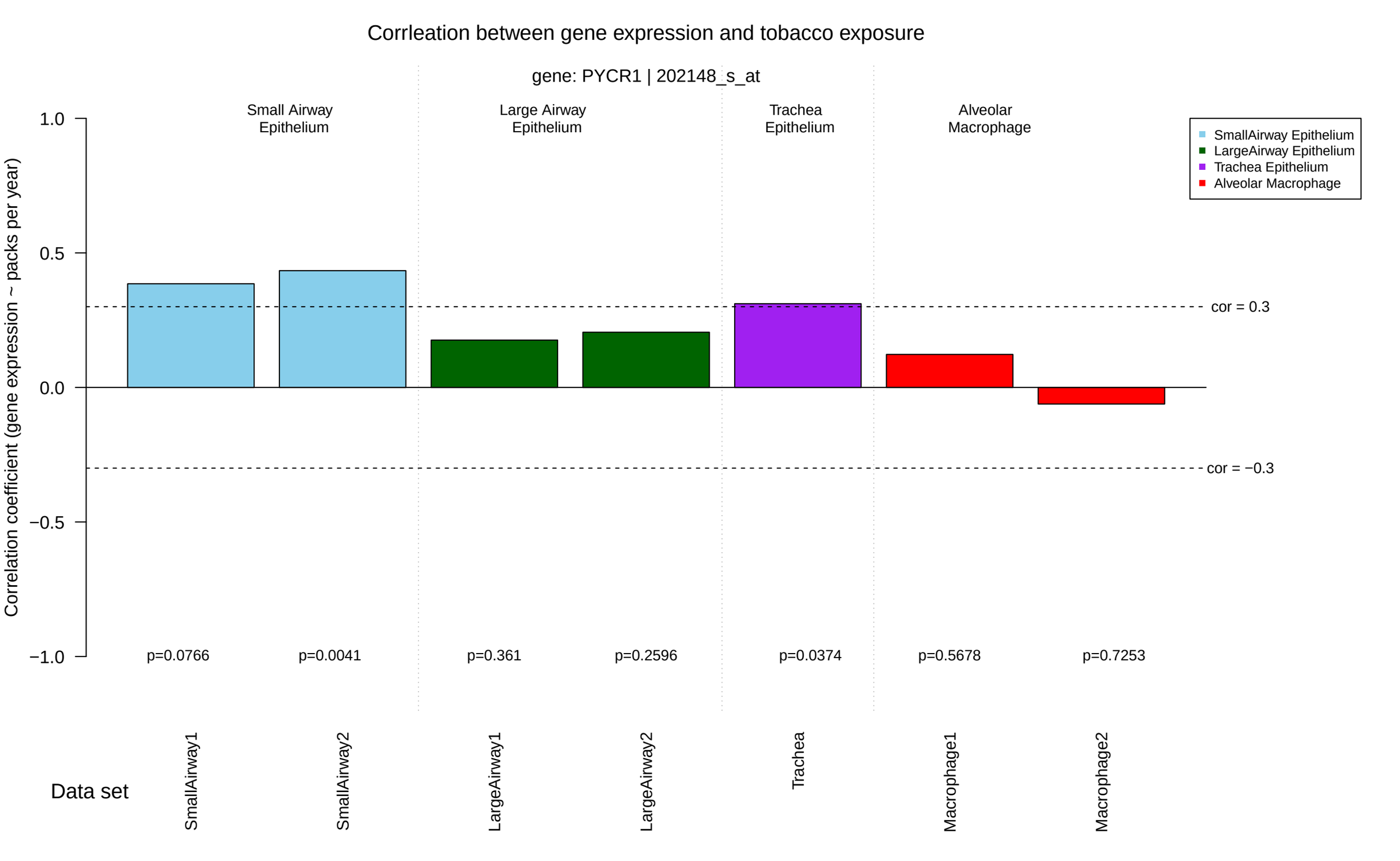

set: LargeAirway1 (p=0.0467)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0989)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1133)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7474)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6668)
PYCR1 | 202148_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PYCR1 | 202148_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0766
p=0.0041
p=0.361
p=0.2596
p=0.0374
p=0.5678
p=0.7253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2